

MAY THE FORGETTING BE WITH YOU: Alternate Replay for Learning with Noisy Labels

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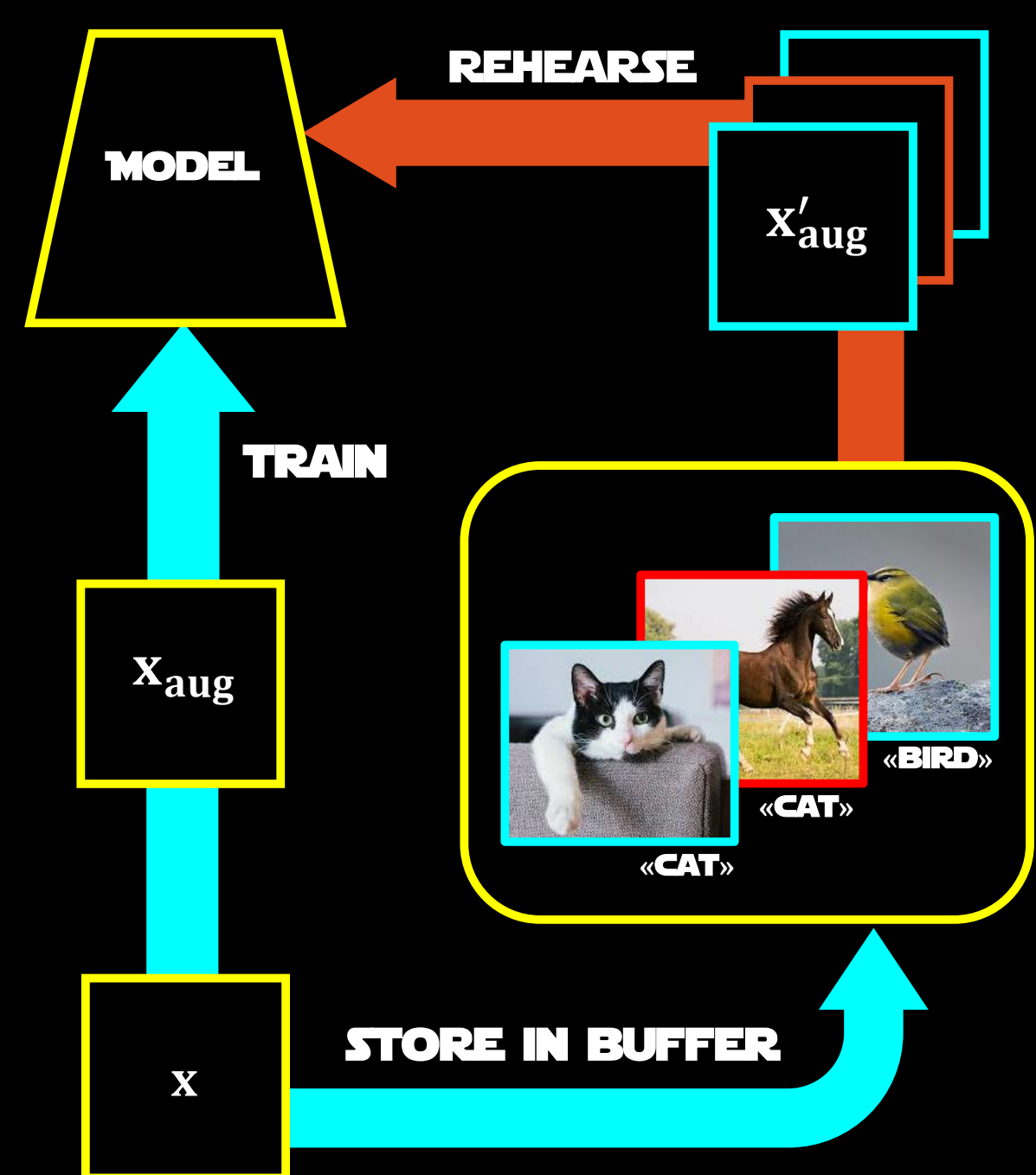
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TL;DR

Alternate Experience Replay (AER) exploits selective forgetting to maintain a clear distinction between the loss of clean and noisy samples. This way, we can spot noisy samples in Continual Learning, improving memory buffer quality (purity and diversity) and performance.

1. Continual Learning under Noisy Labels



Experience Replay (ER): train on data from a sequence of tasks $\{\mathcal{D}_1, \dots, \mathcal{D}_T\}$ with a memory buffer $\mathcal{M}$

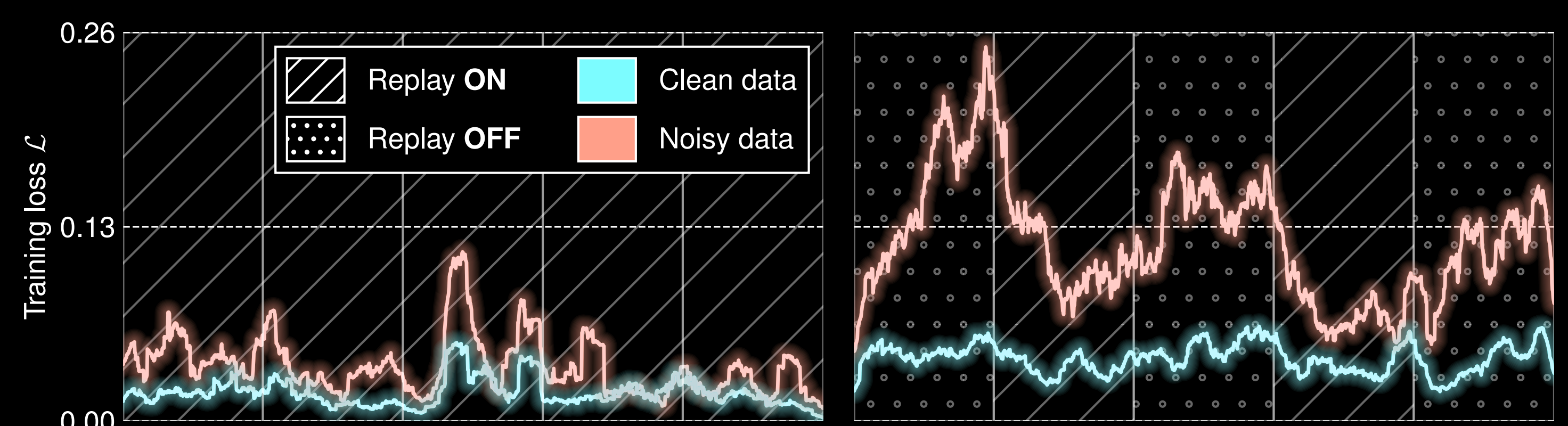
$$\min_{\theta} \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_t} [\mathcal{L}(f(\mathbf{x}), y)] + \mathcal{L}_R$$

$$\mathcal{L}_R = \mathbb{E}_{(\mathbf{x}_r, y_r) \sim \mathcal{M}} [\mathcal{L}(f_{\theta}(\mathbf{x}_r), y_r)]$$

Noisy label space $\tilde{\mathcal{Y}}$ and noisy distribution $\tilde{\mathcal{D}} = \{(\mathbf{x}_i, \tilde{y}_i)\}_{i=1}^N$!

2. Alternate Experience Replay

Exploit DNN's **memorization** effect [1] and **small-loss** criterion to fill the replay memory $\mathcal{M}$ with the clean examples.



Standard replay makes losses indistinguishable (left); alternating epochs of replay and forgetting → significant loss separation (right).

3. Updating the Memory Buffer via Asymmetric Balanced Sampling



Modify *reservoir sampling* to be *loss-aware* [2].

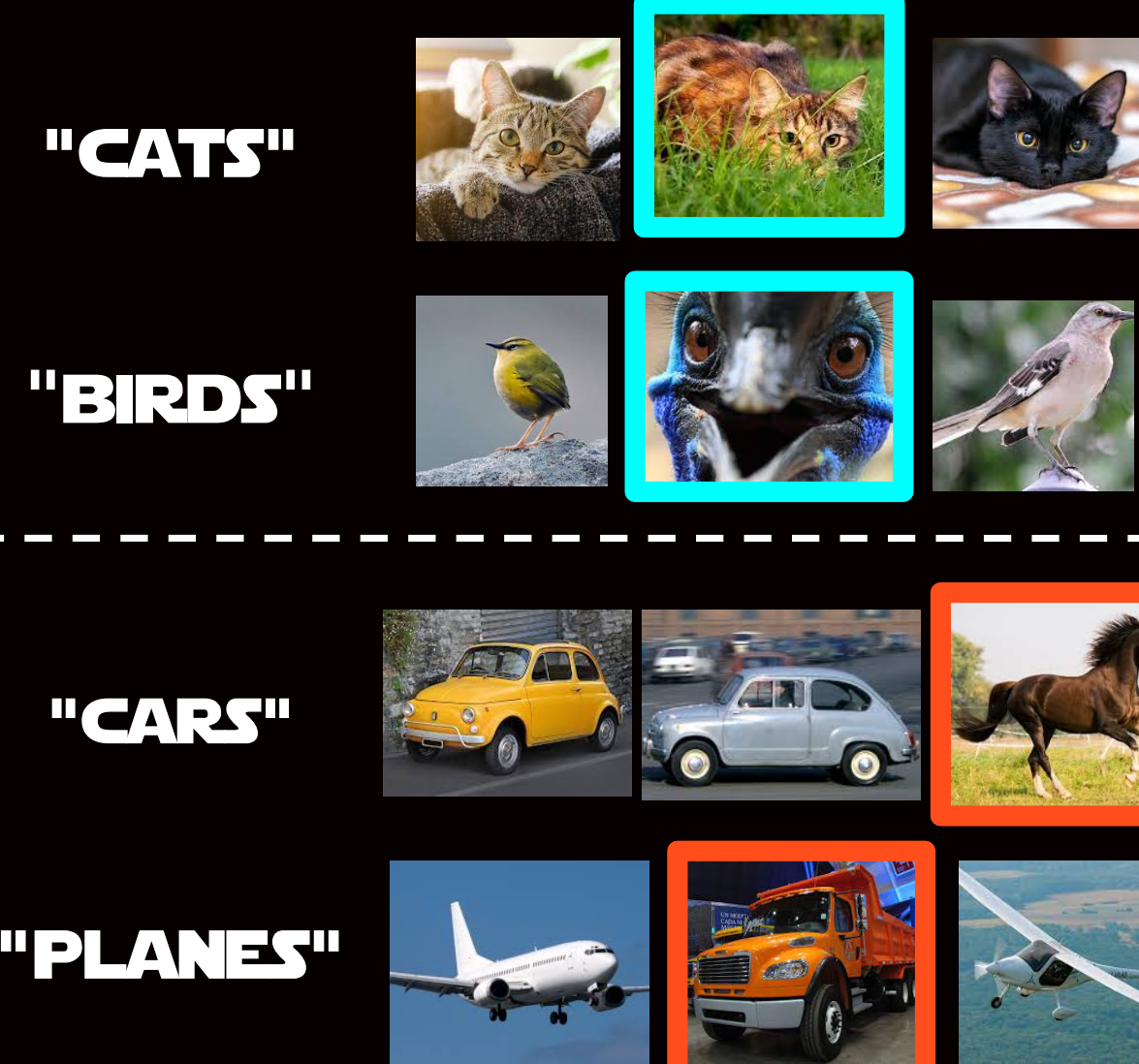
Sample insertion

$$\mathcal{R} = \{(\mathbf{x}, \tilde{y}) \in \mathcal{B} : \mathcal{L}(\mathbf{x}, \tilde{y}) < \mathcal{L}_{\alpha}\}$$

Sample selection

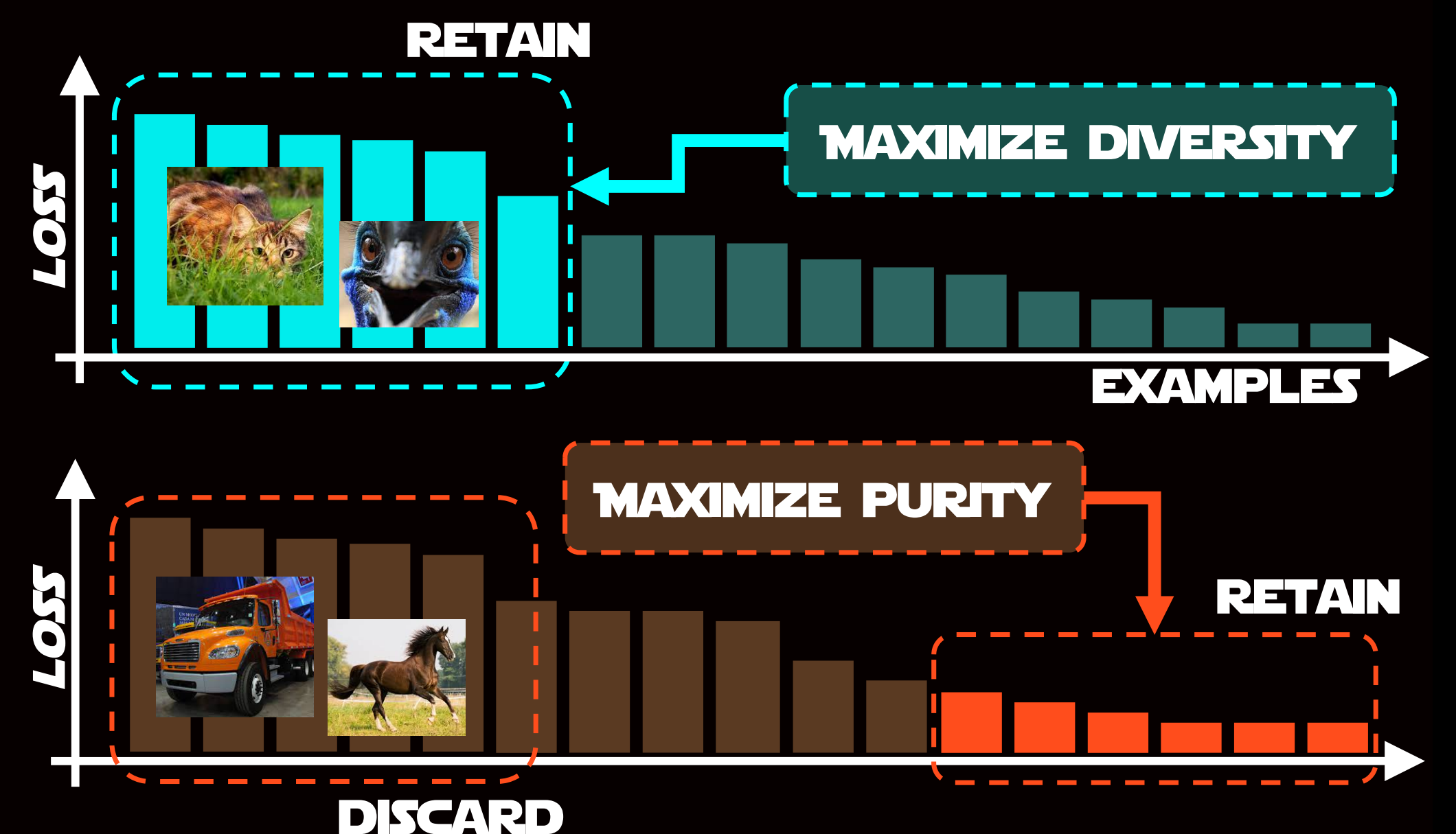
$$s(\mathbf{x}) = \begin{cases} \mathcal{L}(\mathbf{x}, \tilde{y}), & \text{if } (\mathbf{x}, \tilde{y}) \sim \mathcal{D}_t \\ -\mathcal{L}(\mathbf{x}, \tilde{y}), & \text{if } (\mathbf{x}, \tilde{y}) \sim \mathcal{D}_{<t} \end{cases}$$

MEMORY BUFFER



PAST
EXAMPLES

CURRENT
EXAMPLES

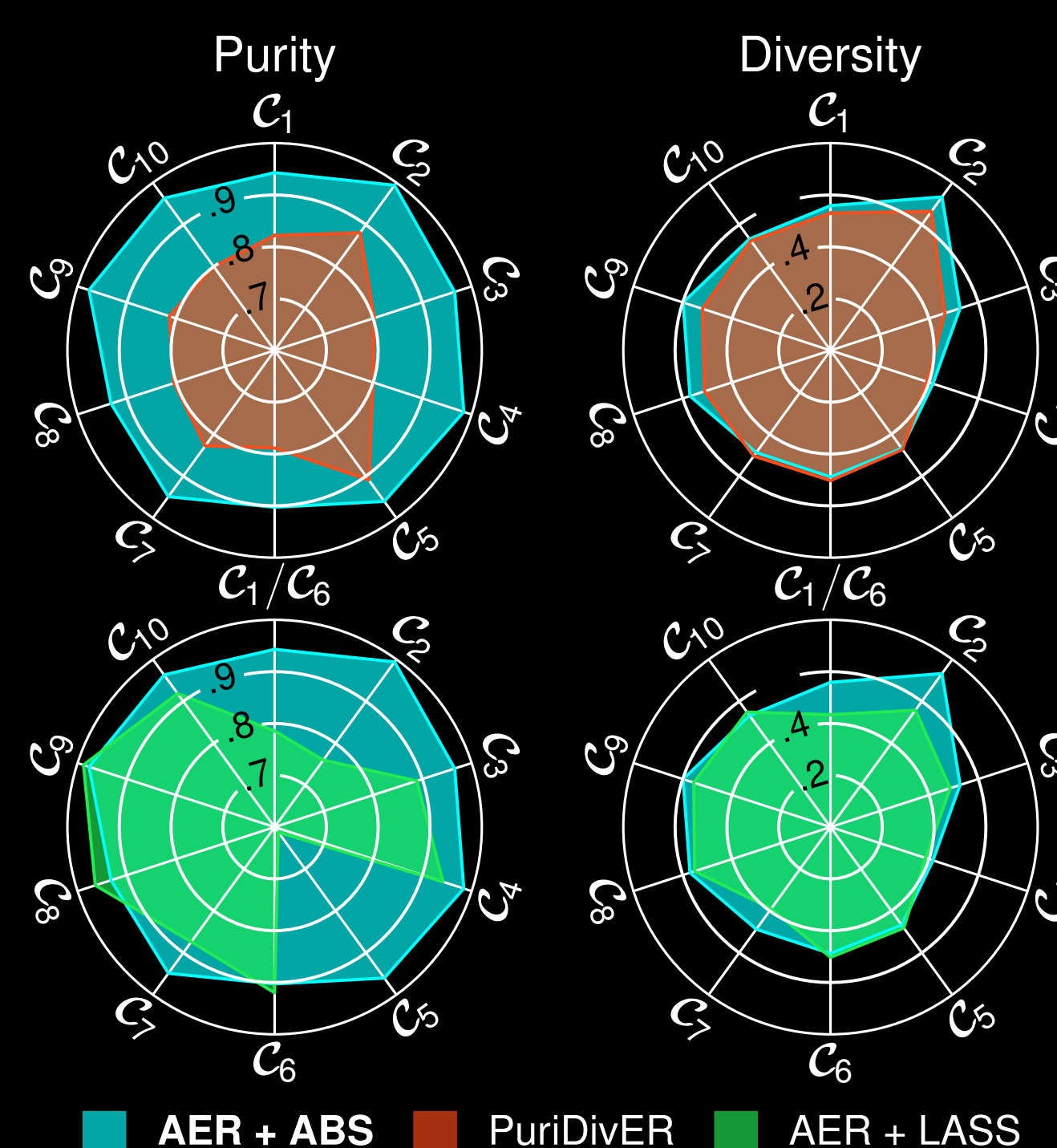


Samples to be released are chosen among the past easier ones, while we reverse the objective for the current task samples to maximize purity.

4. Results

Benchmark	Seq. CIFAR-100				Seq. NTU-60			
		symm		asymm		symm		
Noise rate	20	40	60	20	40	20	40	
Joint	54.77	38.46	23.36	56.70	42.61	68.26	63.02	
Finetune	08.65	07.55	06.15	07.78	05.73	14.30	11.73	
ER [3]	25.14	14.64	8.92	29.42	18.91	29.95	16.02	
+ CoTeaching	25.79	14.46	8.92	32.18	20.76	43.87	30.71	
+ DivideMix	33.31	22.91	13.58	36.98	26.10	40.92	32.07	
PuriDivER.ME	41.25	37.61	27.18	41.65	30.22	43.10	38.07	
OURS	44.34	38.64	26.34	41.24	29.26	47.71	43.11	
w. consolidation	46.11	40.27	34.81	43.67	32.64	48.73	45.19	

5. Analysis

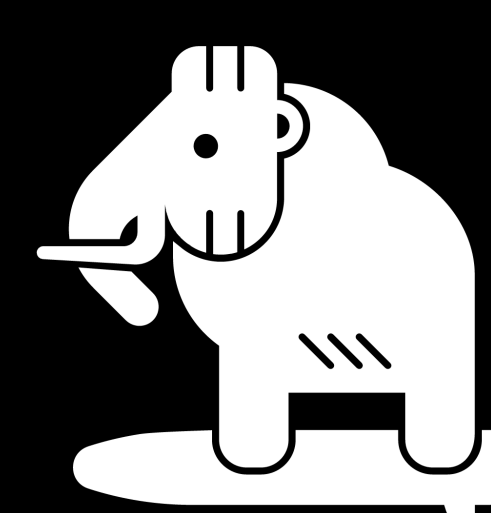


Seq. CIFAR-100 – 60% <i>symm</i>					
ER w.	ACE	α	AER	ABS	FAA
✓	✓				11.65
✓	✓	✓			19.97
✓	✓	✓	✓		24.19
✓	✓	✓		✓	21.68
✓	✓	✓	✓	✓	26.34
✓		✓	✓	✓	22.02

We introduce our main contributions incrementally, with each feature boosting performance.

References

- [1] D. Arpit et al. A closer look at memorization in deep networks. In *International conference on machine learning*, pages 233–242. PMLR, 2017.
- [2] P. Buzzega et al. Rethinking experience replay: a bag of tricks for continual learning. In *International Conference on Pattern Recognition*, 2021.
- [3] J. S. Vitter. Random sampling with a reservoir. *ACM Transactions on Mathematical Software (TOMS)*, 1985.



Mammoth

A PyTorch benchmark library for **Continual Learning** with 20+ datasets and 50+ methods.

